

Summary

Software engineer with **5 years** at **AWS Redshift** building **data warehouse optimization** and **Zero-ETL** streaming replication. Expertise in **distributed storage**, **database internals**, and **production operations at petabyte scale**.

Experience

Amazon Web Services, Redshift

Software Development Engineer II

Dec 2023 - Present

East Palo Alto

- Eliminated **2000+ daily out-of-memory errors** in ALTER TABLE operations through **memory optimization and columnar inserts**, unblocking customer workloads in datasharing scenarios and memory constrained instances
- Built **Aurora PostgreSQL to Redshift Zero-ETL** integration, enabling **real-time data replication** without transformation pipelines for hundreds of customers
- Implemented checkpoint caching mechanism for Zero-ETL, **reducing recovery time to minutes from multiple days** in case of cluster restarts and resumes
- Migrated **2,000+ integration and system tests** from Python 2 to Python 3, **modernizing team testing infrastructure** while mentoring junior engineers
- AWS Security Guardian** for the Redshift Data Management Team

Amazon Web Services, Redshift

Software Development Engineer I

July 2021 - Dec 2023

East Palo Alto

- Delivered **automatic table distribution** system during ingestion, **improving query performance by 20%** fleet-wide with minimal ingestion overhead
- Automated maintenance prioritization** for **10,000+ tables** daily using **intelligent scoring** based on **query patterns** and **resource constraints**

Aikaan Labs (Acquired by Ather Energy)

Network Analytics Engineer

July 2018 - June 2019

Bengaluru

- Designed** and **built** an **edge machine learning system** that employed **information theory** and **neural nets** to **predict** edge device health, allowing a Raspberry Pi edge server to handle **50,000+ edge devices**
- Automated** the **machine learning stack**, leading to a **3300% increase** in **prediction efficiency**
- Developed SQL parsers** and **telemetry collectors** in Go for dashboarding and system analytics

Education

University of Southern California

Master of Science, Computer Science

3.6 GPA

August 2019 - May 2021

- Courses:** Foundations of Data Management, Database Mangement Systems, Analysis of Algorithms, Foundations of AI, Machine Learning
- Societies:** USC ACM

REVA University

Bachelor of Technology, Computer Science and Engineering

3.76 GPA

August 2014 - June 2018

- Courses:** Machine Learning, Database Management Systems, Operating Systems
- Societies:** Co-Founded GLUG-REVA (GNU/Linux User Group)

Projects

Tensor Cortex

<https://github.com/Skeletrox/tensor-cortex>

March 2020

- Distributed, Containerized Deep Learning System

Musiquity

<https://github.com/Skeletrox/musiquity>

November 2020

- Unified analytics engine that used time-series data, NoSQL, and relational data to recommend a workout mix for the user through Machine Learning and recommendation algorithms

Skills

C++	Python	Rust	Database Internals
Distributed Systems	ML/AI	Java	ETL

Certifications

Oracle Certified Associate in Java Programming

Oracle

<https://www.credly.com/badges/f2fa6ea5-64d6-4c84-92d8-457b78cacf25>

March 2017